HOPE LAB / METHOD REFERENCE

LAB Autoaggregation and Coaggregation Assays

A literature-based reference to aggregation concepts, data handling, controls and interpretation.

HOPE Lab student reference. This document is maintained as a convenient reference for HOPE Lab students and members. It is not an approved institutional SOP and does not replace supervisor instruction, biosafety review, chemical risk assessment, instrument-specific training, or current institutional procedures. Wet-lab work must be conducted only under an approved, supervisor-directed protocol using appropriately assessed materials and facilities.

Purpose and scientific principle

Autoaggregation describes the time-dependent association of cells from the same strain. Coaggregation describes the association observed when two different microbial partners are combined. Optical-density change in the non-sedimented fraction is commonly used as an indirect measure. These assays are phenotypic screening tools; they do not directly prove epithelial adhesion, colonization, or probiotic efficacy.

Analysis workflow (non-operational)

- Use only authenticated, supervisor-approved strains and the current institutional method for any live-cell preparation or assay handling.
- Define the comparison groups before analysis: each strain alone, the paired combination, and appropriate matrix/background controls.
- Record the baseline optical-density value and the corresponding value at each method-defined observation point, using the same instrument settings and blanking approach.
- Calculate autoaggregation separately for each monoculture. Calculate coaggregation from the two monoculture values and the paired-mixture value, following the formula specified by the chosen publication or SOP.
- Use independent biological replicates and report the number of replicates, summary statistic, variability, and statistical test.
- Interpret aggregation as an in-vitro surface/association phenotype only. Do not equate a high percentage with proven host adhesion or health benefit.

Common calculations

A commonly used autoaggregation expression is: Autoaggregation (%) = $[(A_0 - A_t) / A_0] \times 100$, where A_0 is the initial optical density and A_t is the value at the defined later observation point.

A commonly used coaggregation expression is: Coaggregation (%) = $\{[(A_x + A_y) / 2] - A_{mix}\} / [(A_x + A_y) / 2] \times 100$, where A_x and A_y are the monoculture values and A_{mix} is the paired-mixture value at the same defined observation point.

Formula conventions differ among publications. State the exact equation, time point, wavelength, blank correction, and whether the calculation uses initial or time-matched monoculture values. Do not compare percentages generated using different definitions.

Controls and quality checks

- Matrix/background blank appropriate to the measurement system.
- Each partner measured separately for coaggregation calculations.

- A validated comparison strain or reference condition, where available and justified by the study.
- Consistent optical path, instrument settings, sample handling and time-point definitions as specified in the approved SOP.
- Replicate agreement and documentation of visible sedimentation or clumping that could distort optical measurements.

Limitations

Optical-density decline can reflect sedimentation, aggregation, cell-size differences, or other physical effects. A single endpoint can hide kinetics. Aggregation is not equivalent to adhesion to host tissue; follow-up experiments and independent evidence are required for stronger biological claims.

References and authoritative resources

1. Kos B, Šuškovi■ J, Vukovi■ S, Šimpraga M, Frece J, Matoši■ S. Adhesion and aggregation ability of probiotic strain Lactobacillus acidophilus M92. Journal of Applied Microbiology. 2003;94(6):981-987. doi:10.1046/j.1365-2672.2003.01915.x.

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